

Optimal Classification Tree Case Studies

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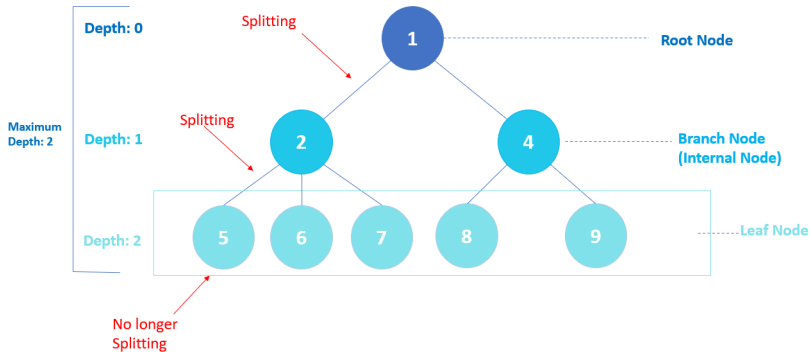
Reference Paper: *Optimal Classification Trees*, Bertismas and Dunn

Outline

- 1 Classification and Regression Trees (CART)
 - Tree
 - Binary Tree
 - Classification and Regression Trees (CART)
- 2 Optimal Classification Tree (OCT)
 - Introduce OCT
 - Problem Formulation of OCT
 - Numerical Experiments
- 3 Results & Conclusion
- 4 Improvements

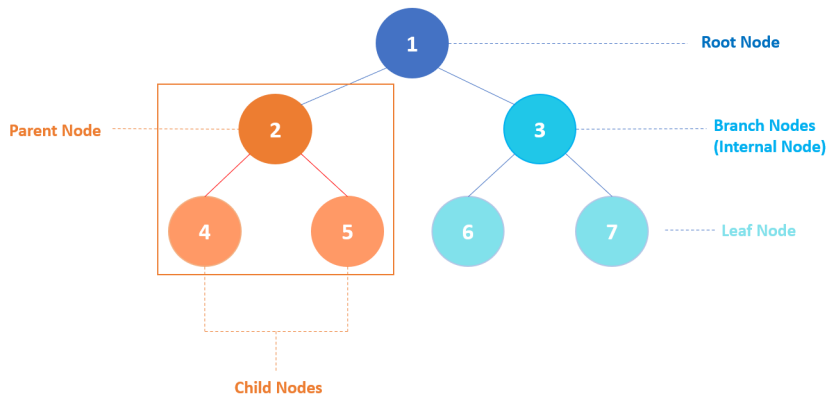
Tree

- What is a Tree?



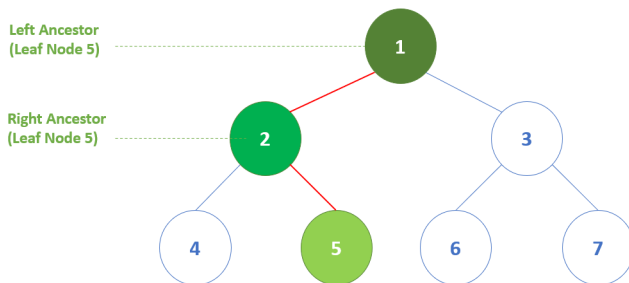
Binary Tree

- What is a Binary Tree? A classic type of tree
 - ▶ Maximum number of nodes: $2^{(D+1)} - 1$
 - ▶ Maximum number of leaf nodes: 2^D
- What is a Parent Node & a Child Node?



Binary Tree - continue

- What is Left Ancestor & Right Ancestor?

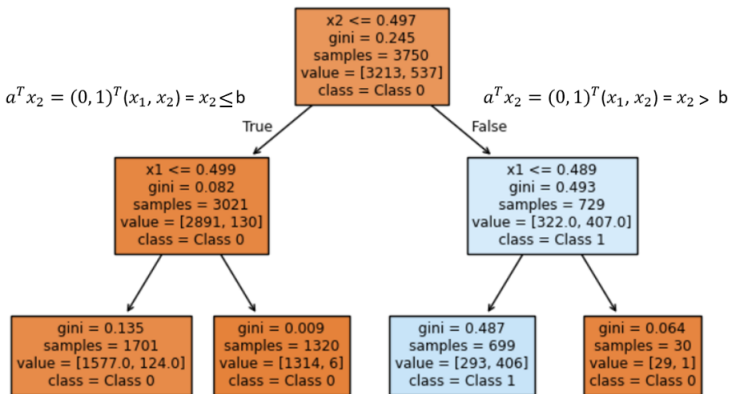


Classification and Regression Trees (CART)

- What is a Decision Tree (Classification Tree)?
- What is Classification and Regression Trees (CART)?

Generate a CART by using Gini impurity:

CART Tree Structure - matrix_train (max_depth=2)



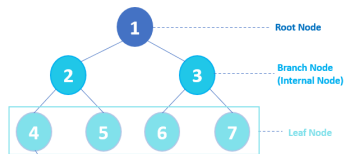
CART - continue

- (x_i, y_i) : n observations (x_i, y_i) , $i = 1, \dots, n$, each with p features $x_i \in [0, 1]^p$, and a class label $y_i \in \{1, \dots, K\}$
- T : Decision tree
- α : Alpha (α) is a complexity penalty parameter that controls the trade-off between model accuracy and simplicity in decision tree algorithms. It penalizes the size of the tree to prevent overfitting. (see graphs in next slides)
- $N_x(\ell)$: Number of training points contained in leaf node ℓ
- $N_{\min}$: Minimum number of points required in any leaf node
- $R_{xy}(T)$: Misclassification error of the tree T on training data
- $|T|$: Number of branch nodes
- The objective of CART is given by:

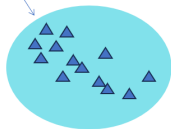
$$\min \quad R_{xy}(T) + \alpha |T| \quad \text{s.t.} \quad N_x(\ell) \geq N_{\min} \quad \forall \ell \in \text{leaves}(T) \quad (1)$$

Overfitting & Underfitting

- What is overfitting & underfitting?
 - ▶ Overfitting is when a model learns the training data too well, leading to the model not working well on new data.
 - ▶ Underfitting is when a model is too simple to capture the features, leading to it not performing well enough even on the training data.
- How to prune an overfitting tree?

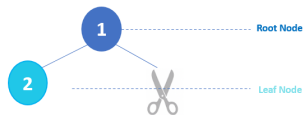


Leaf Node 4

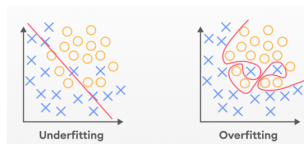


■ Misclassified points
▲ Correctly classified points

Pruning the Tree



Prevent Overfitting



CART - continue

- CART relies on Greedy top-down heuristics to produce good splits at each node, fails to consider the tree's global optimality
- Other popular methods, such as Neural Network and Random Forest, are difficult or impossible to explain which features influence their decision
- How to improve the CART?

Introduction to Optimal Classification Tree (OCT)

- In Optimal Classification Tree (Bertsimas and Dunn 2017), Bertsimas and Dunn introduced the Optimal Classification Trees (OCT), a new formulation of the decision tree problem that produces optimal decision trees with univariate splits.
- The OCT has the same objective as CART and is stated as a formal mixed integer optimization (MIO) problem and it aims at minimizing the decision tree's overall misclassification rate.

Introduction to OCT - continue

- To structure a tree as an MIO problem, one must consider the following decision making rules.
 - ▶ At each new branch node, OCT must choose to split or to stop;
 - ▶ If the node chooses to split, then a feature must be chosen to split on;
 - ▶ If node chooses not to split, a label must be chosen for the resulting leaf node;
 - ▶ When classifying the training set data points according to the tree under construction, we must assign points to leaf nodes such that the tree structure is preserved.

Problem Formulation of OCT

- Let D be the depth of the tree;
- let $\{(X, y) \mid X \in \mathbb{R}^{n \times p}, y_i \in \mathbb{N} \cup \{0\}\}$ be the classification dataset of n observations and p features, where y_i is the class label of the i^{th} observation represented by integers.
- let each feature be normalized such that $x_{ij} \in [0, 1]$
 $\forall i = 1, \dots, n, \quad j = 1, \dots, p;$
- let $\mathcal{T}_B = \{1, 2, \dots, \lfloor T/2 \rfloor\}$ be the set of branch nodes, and let $\mathcal{T}_L = \{\lfloor T/2 \rfloor + 1, \dots, T\}$ be the set of leaf nodes, where $T = 2^{(D+1)} - 1$ is the number of nodes in a maximum binary tree;
- let $A_L(t)$ denotes the set of left ancestors of a node t , and let $A_R(t)$ denotes the set of right ancestors of a node t ;
- let $p(t)$ denotes the parent node of node t .

Problem Formulation of OCT

- Let the splitting decision and criterion on the branch nodes be tracked with variables a , b , and d with constraints (2), (3), and (4). Constraint (5) enforces the hierarchical tree structure by preventing an orphan node.

$$a = \left\{ a_t \in \mathbb{R}^p, t \in \mathcal{T}_B \mid \sum_{j=1}^p (a_{tj}) = 1 \right\},$$

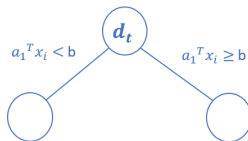
$$b = \{b_t \in [0, 1], t \in \mathcal{T}_B\}, d = \{\mathbf{1}\{\text{node } t \text{ applies a split}\}, t \in \mathcal{T}_B\}.$$

f: feature

	f1	f2	f3	...	ft
a_{1t}	a_{11}	a_{12}	a_{13}	...	a_{1t}
a_{2t}	a_{21}	a_{22}	a_{23}	...	a_{2t}
a_{3t}	a_{31}	a_{32}	a_{33}	...	a_{3t}
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
a_{jt}	a_{j1}	a_{j2}	a_{j3}	...	a_{jt}

$d_t = 1$ (blue box around the first row)

$d_t = 0$ (red box around the rest of the rows)



Problem Formulation of OCT - Continue

- Let the splitting decision and criterion on the branch nodes be tracked with variables a , b , and d with constraints (2), (3), and (4). Constraint (5) enforces the hierarchical tree structure by preventing an orphan node.

$$\sum_{j=1}^p (a_{tj}) = d_t, \quad \forall t \in \mathcal{T}_B \quad (2)$$

$$0 \leq b_t \leq d_t, \quad \forall t \in \mathcal{T}_B \quad (3)$$

$$a_{tj} \in \{0, 1\} \quad (4)$$

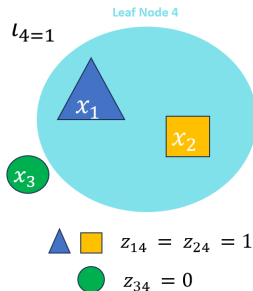
$$d_t \leq d_{p(t)}, \quad \forall t \in \mathcal{T}_B \setminus \{1\} \quad (5)$$

Problem Formulation of OCT - continue

- Let the allocation of a point to the leaf nodes be tracked by the indicator variables z and l .
- We enforce the minimum number of points in each leaf to be N_{min} and that no point can be assigned to more than one leaf.

$$z = \{z_{it} = \mathbf{1}\{x_i \text{ is in node } t\}, t \in \mathcal{T}_L, i = 1, \dots, n\},$$

$$l = \{l_t = \mathbf{1}\{\text{leaf } t \text{ contains any points}\}, t \in \mathcal{T}_L\}.$$



$$z_{it} \leq l_t, \quad \forall t \in \mathcal{T}_L \quad (6)$$

$$\sum_{i=1}^n z_{it} \geq N_{min} l_t, \quad \forall t \in \mathcal{T}_L \quad (7)$$

$$\sum_{t \in \mathcal{T}_L} z_{it} = 1, \quad i = 1, \dots, n \quad (8)$$

Problem Formulation of OCT - continue

- We need enforce the splits that are required by the structure of the tree when assigning points to leaves. If leaf node t contains point x_i , then:
 - ▶ For all $l \in A_L(t)$, the split must satisfy: $a_l^\top x_i < b_l$ (9).
 - ▶ For all $r \in A_R(t)$, the split must satisfy: $a_r^\top x_i \geq b_r$ (10).
- Since MIO solvers do not support strict inequalities, a small vector ϵ_j is added to x_i in (9). We specify an ϵ_j for each feature j and let $\mathbf{x}_j$ denotes the sorted j th feature, then

$$\epsilon_j = \min \left\{ \mathbf{x}_j^{(i+1)} - \mathbf{x}_j^{(i)} \mid \mathbf{x}_j^{(i+1)} \neq \mathbf{x}_j^{(i)}, i = 1, \dots, n \right\}, \quad j = 1, \dots, p$$

$$\epsilon_{\max} = \max_j \{\epsilon_j\}$$

$$\begin{aligned} a_l^\top (x_i + \epsilon) &\leq b_l + (1 + \epsilon_{\max})(1 - z_{it}), \\ i &= 1, \dots, n, \quad \forall t \in \mathcal{T}_L, \quad \forall l \in A_L(t) \end{aligned} \quad (11)$$

$$a_r^\top x_i \geq b_r - (1 - z_{it}), \quad i = 1, \dots, n, \quad \forall t \in \mathcal{T}_L, \quad \forall r \in A_R(t). \quad (12)$$

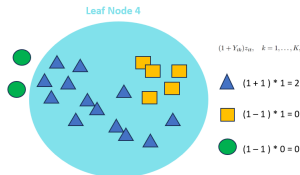
Problem Formulation of OCT - continue

- Let $k = 1, \dots, K$ indicate the class label represented by integers, and let the Y matrix track the class label y_i of each point. We track the number of points of label k in leaf node t with variables N_{kt} .

$$Y_{ik} = \begin{cases} +1, & \text{if } y_i = k \\ -1, & \text{otherwise} \end{cases}, \quad k = 1, \dots, K, \quad i = 1, \dots, n$$

$$N_{kt} = \frac{1}{2} \sum_{i=1}^n (1 + Y_{ik}) z_{it}, \quad k = 1, \dots, K, \quad \forall t \in \mathcal{T}_L \quad (13)$$

$$N_t = \sum_{i=1}^n z_{it}, \quad \forall t \in \mathcal{T}_L \quad (14)$$



Problem Formulation of OCT - continue

- Let c_t be the variables that track the predicted class label, which is the most popular class among the leaf t . We restrict each leaf node to have only one class prediction by introducing c_{kt} and constraint (16).

$$c_t = \arg \max_{k=1,\dots,K} \{N_{kt}\}, \quad \forall t \in \mathcal{T}_L \quad (15)$$

$$c_{kt} = \mathbf{1}\{c_t = k\}, \quad \forall k = 1, \dots, K, \quad \forall t \in \mathcal{T}_L$$

$$\sum_{k=1}^K c_{kt} = I_t, \quad \forall t \in \mathcal{T}_L \quad (16)$$

Problem Formulation of OCT - continue

- Finally, let L denote the misclassification loss for the leaf nodes t :

$$L_t = N_t - \max_{k=1,\dots,K} \{N_{kt}\}, \quad (17)$$

$$L_t \geq N_t - N_{kt} - n(1 - c_{kt}), \quad k = 1, \dots, K, \quad \forall t \in \mathcal{T}_L, \quad (18)$$

$$L_t \leq N_t - N_{kt} + nc_{kt}, \quad k = 1, \dots, K, \quad \forall t \in \mathcal{T}_L, \quad (19)$$

$$L_t \geq 0, \quad \forall t \in \mathcal{T}_L, \quad (20)$$

Objective of OCT

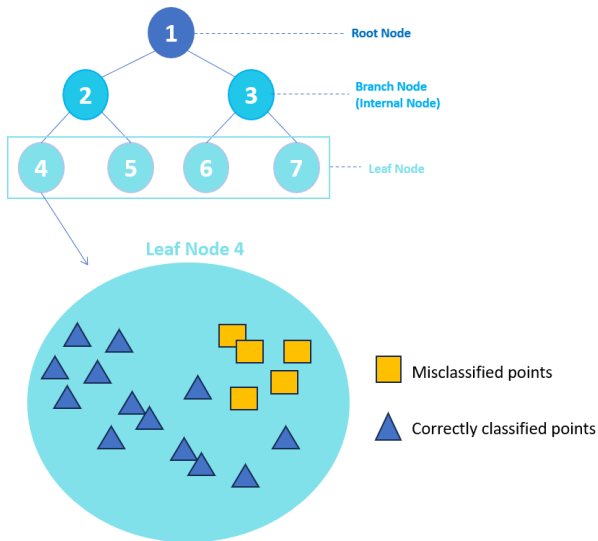
- Let $\hat{L}$ be the baseline accuracy of the entire dataset:

$$\hat{L} = \frac{\text{Number of points of the most popular class of the entire dataset}}{n = \text{Total number of observations}}$$

- and let α be the complexity parameter.
- The Objective of OCT is given by:

$$\min \frac{1}{\hat{L}} \sum_{t \in \mathcal{T}_L} L_t + \alpha \sum_{t \in \mathcal{T}_B} d_t$$

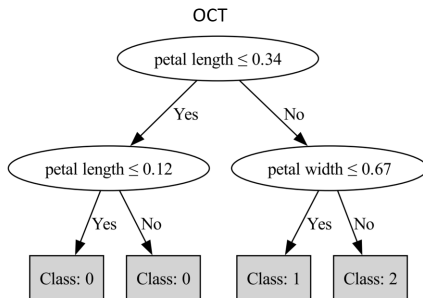
Example of Missclassified points in Node 4



Numerical Experiments

- 1 Set $D = 2$; $\alpha = 0.5$; $d_t = 1 \forall t \in \mathcal{T}_L$; $N_{min} = \lfloor 0.05n \rfloor$, where n is the number of observations.
- 2 Use 4-fold cross-validation for each dataset, training the OCT on three folds and testing on the remaining validation set.
- 3 Define accuracy by:

$$\text{Accuracy} = \frac{\sum_{t \in \mathcal{T}_L} (\text{Points in Leaf Node } t - \text{Misclassified Points in Leaf Node } t)}{\text{Total Number of Observations}}$$



Experiment Results - Overall

Table 1 Result

Data Sets	Blood	Iris	Heart Disease	Wine	Soybean-small	Spect-heart	Land Mines	Connectionist	Australian Credit	Fertility	matrix_train
	id = 176	id = 53	id = 45	id = 109	id = 91	id = 95	id = 763	id = 151	id = 143	id = 244	txt.
# Observations (before sampling)	748	150	303	178	47	267	338	208	690	100	304372
#Observations for Training	748	150	297	178	47	267	338	100	690	100	5000
# Features	4	4	13	13	35	22	3	60	14	9	2
# Classes	2	3	5	3	4	2	5	2	2	2	2
Tree Depth	2	2	2	2	2	2	2	2	2	2	2

CART

Train Set Accuracy	0.762	0.962	0.6	0.936	0.83	0.801	0.491	0.833	0.8685	0.88	0.895
Test Set Accuracy	0.753	0.946	0.532	0.859	0.682	0.757	0.432	0.59	0.864	0.88	0.893

OCT

AVG Runtime (seconds)	96	6	3000	30	0.2	15	180	6000	4680	40	3273
Gap (%)	0	0	0	0	0	0	0	56.82	0	0	0
Train Set Accuracy	0.776	0.962	0.614	0.97	1	0.801	0.524	0.903	0.866	0.88	0.899
Test Set Accuracy	0.742	0.946	0.528	0.933	1	0.757	0.474	0.75	0.846	0.88	0.898

XGBoost

Train Set Accuracy	0.839	1	0.996	1	1	0.911	0.907	1	1	0.98	0.928
Test Set Accuracy	0.766	0.946	0.545	0.966	0.958	0.787	0.53	0.73	0.882	0.86	0.899

Random Forest

Train Set Accuracy	0.768	0.964	0.602	1	1	0.794	0.525	1	1	0.88	0.998
Test Set Accuracy	0.769	0.946	0.559	0.977	1	0.794	0.346	0.72	0.88	0.88	0.888

depth = 2

Experiment Results - CART VS. OCT

Table 2 Result (CART VS. OCT):

Data Set				Testing set accuracy		Improvement
				CART	OCT	OCT-CART
Blood	748	4	2	0.753	0.742	-1.10%
Iris	150	4	3	0.946	0.946	0.00%
Heart Disease	297	13	5	0.532	0.528	-0.40%
Wine	178	13	3	0.859	0.933	7.40%
Soybean-small	47	35	4	0.682	1	31.80%
Spect-heart	267	22	2	0.757	0.757	0.00%
Land Mines	338	3	5	0.432	0.474	4.20%
Connectionist	100	60	2	0.59	0.75	16.00%
Australian Credit Approval	690	14	2	0.864	0.846	-1.80%
Fertility	100	9	2	0.88	0.88	0.00%
Matrix_Train	5000	2	2	0.893	0.898	0.50%

depth=2

Table 3

IF OCT win			
OCT Accuracy(%)	CART win	OCT win	AVG. Accuracy improvement (%)
0-60	1	1	4.20%
60-70	0	0	0.00%
70-80	1	2	8.00%
80-90	1	2	0.25%
90-100	0	3	13.07%
Total	3	8	

Experiment Results - XGBoost VS. OCT

Table 4 Result (XGBoost VS. OCT):

Data Set				Testing set accuracy		Improvement
				XGBoost	OCT	OCT-XGBoost
Blood	748	4	2	0.766	0.742	-2.40%
Iris	150	4	3	0.946	0.946	0.00%
Heart Disease	297	13	5	0.545	0.528	-1.70%
Wine	178	13	3	0.966	0.933	-3.30%
Soybean-small	47	35	4	0.958	1	4.20%
Spect-heart	267	22	2	0.787	0.757	-3.00%
Land Mines	338	3	5	0.53	0.474	-5.60%
Connectionist	100	60	2	0.73	0.75	2.00%
Australian Credit Approval	690	14	2	0.882	0.846	-3.60%
Fertility	100	9	2	0.86	0.88	2.00%
Matrix_Train	5000	2	2	0.899	0.898	-0.10%

depth=2

Table 5

IF OCT win

OCT Accuracy(%)	IF OCT win		AVG. Accuracy improvement (%)
	XGBoost win	OCT win	
0-60	1	0	0.00%
60-70	0	0	0.00%
70-80	2	1	2.00%
80-90	2	1	2.00%
90-100	1	2	2.10%
Total	7	4	

Experiment Results - Random Forest VS. OCT

Table 6 Result (Random Forest VS. OCT):

				Testing set accuracy		Improvement
Data Set						
Name	n	p	K	Random Forest	OCT	OCT-Random Forest
Blood	748	4	2	0.769	0.742	-2.70%
Iris	150	4	3	0.946	0.946	0.00%
Heart Disease	297	13	5	0.559	0.528	-3.10%
Wine	178	13	3	0.977	0.933	-4.40%
Soybean-small	47	35	4	1	1	0.00%
Spect-heart	267	22	2	0.794	0.757	-3.70%
Land Mines	338	3	5	0.346	0.474	12.80%
Connectionist	100	60	2	0.72	0.75	3.00%
Australian Credit Approval	690	14	2	0.88	0.846	-3.40%
Fertility	100	9	2	0.88	0.88	0.00%
Matrix_Train	5000	2	2	0.888	0.898	1.00%

depth=2

Table 7

			IF OCT win
OCT Accuracy(%)	RF win	OCT win	AVG. Accuracy improvement (%)
0-60	1	1	12.80%
60-70	0	0	0.00%
70-80	2	1	3.00%
80-90	1	2	0.50%
90-100	1	2	0.00%
Total	5	6	

Conclusion

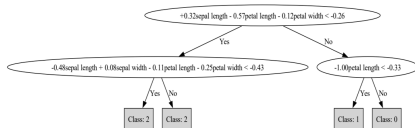
In general, OCT is more likely to outperform CART and even in the case where it loses against CART, the loss is minimal. The OCT has comparable accuracy with Random Forest and XGBoost.

Overall, the result provides comprehensive evidence that OCT has interpretability with each split clearly traceable to specific features and its thresholds, and it leads to significant improvement over heuristic methods.

Improvements

- Adjust the Maximum depth D
- Adjust the complexity parameter Alpha α
- Optimal Classification Trees with Hyperplanes (OCT-H)
- *Strong Optimal Classification Trees* by Aghaei, Gómez, and Vayanos

OCT - H



Max.depth	CART (%)	OCT (%)	OCT - H (%)
1	71.0	71.3	76.1
2	76.2	78.0	81.0
3	78.8	79.5	82.3
4	79.8	80.4	82.9

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